INFORMATICS INSTITUTE OF TECHNOLOGY

UNIVERSITY OF WESTMINSTER

5BUIS0170C – BIS Design and Architecture

Group Coursework Report

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i. Introduction

This completed group coursework report shows a detailed software design analysis for a global platform known as "Foodiverse". Browse, share, and take part in recipes from diverse cultures and cuisines is enabled for its users. The "Foodiverse" platform holds various user roles, ranging from guests to distinguished members, and each level is provided with stepwise enhanced features based on engagement levels.

As the first task, the Class Diagram Design gives a well structured blueprint of the system, gathering boundaries, control, and entity classes. The detailed design class diagram shows the relationships between main components, including user roles of the platform, recipe management of the platform, and profile customisation features. Together, these things provide the foundation for system implementation.

As the second task, the well designed State Machine diagram represents a dynamic behaviour of user progression within the "Foodiverse" platform. This state machine diagram models how the users of this platform transition between various states- from Guest to registered user, contributor, and finally distinguished member- based on triggers, events, actions, and ensuring a clear understanding of the user lifecycle.

As the fourth task, GUI, known as Graphical User Interface design, translates the structural and functional designs into a user centred interface specification. This user interface section mainly focuses on the recipe upload page for registered users, detailing content organisation of the interface, accessible colour schemes, to web design standards and conventions. These are supported by academic matters.

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1. Question On Class Diagram

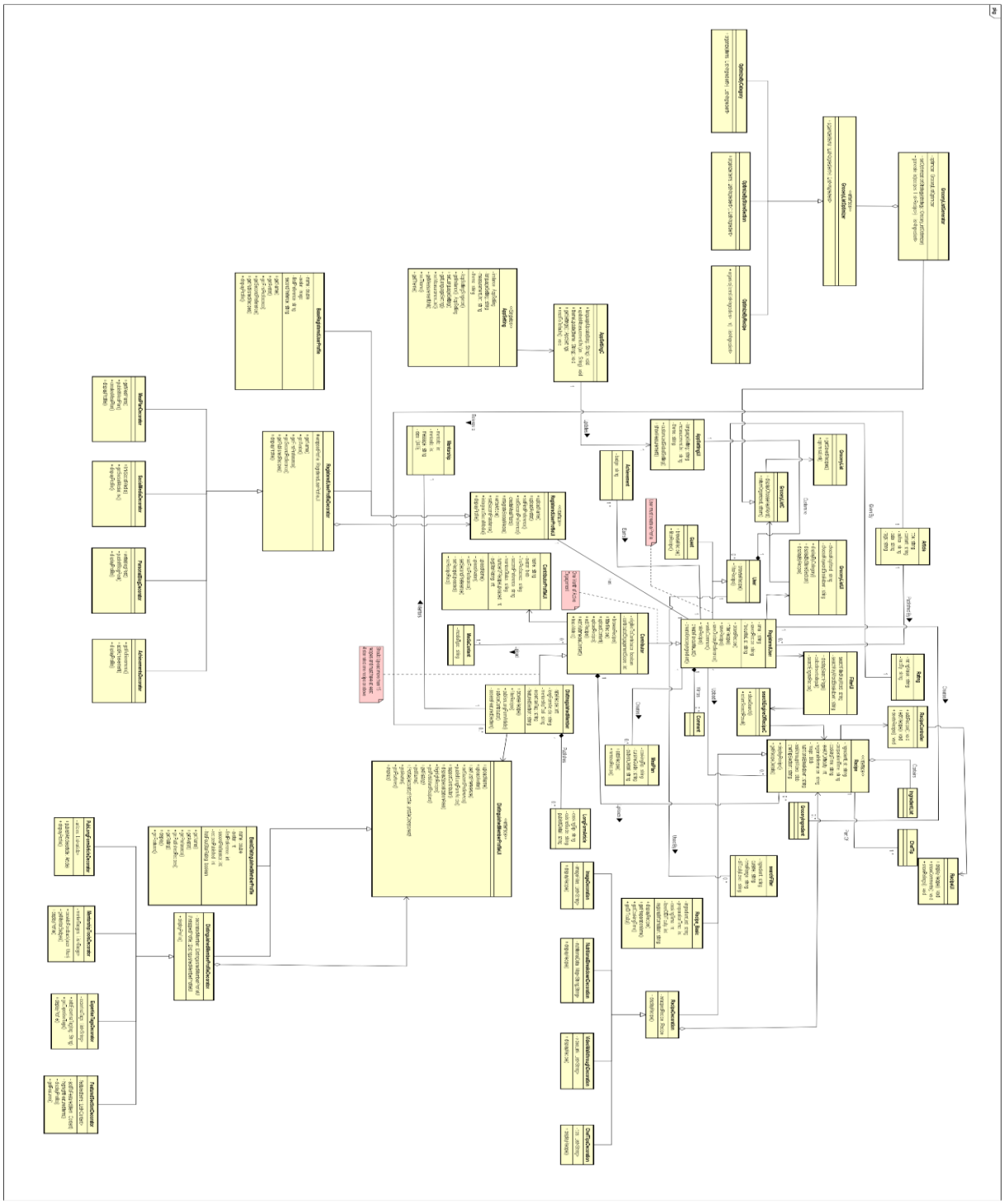

Figure 1 Class Diagram

1.1. Class Diagram Explanation (300 words)

This class diagram, which was created by our group, clearly models the structure of the Foodiverse platform. Identifying more than twenty different types of classes and architecting using the Entity control boundary pattern for logical separation.

The above-mentioned class diagram of the Foodiverse global recipe sharing platform organises its features into interconnected classes that represent users, content, personalisation, and system-wide functionality. In this diagram, there are some generalisations. They are the User superclass, which generalises into Guest and RegisteredUser. There is a part of the diagram that represents the grocery list feature, showing how the saved recipes are converted into an organised shopping list for the user. The GroceryList provides the stored recipes, while the GroceryListC acts as the controller class of this diagram that processes and organises the ingredients, and the GroceryListUI acts as the boundary class which displays the final list in different formats, such as by category, by store section, and by recipe. The associations indicate that the user interacts mainly with the UI, while the controller handles the logic, and the GroceryList class supplies the data.

The RegisteredUser is the base interactive user, and this user can save recipes, save dietary preferences, write comments, rate recipes, create a favourite list, and create a grocery ingredient. The Contributor is a specialised form of the RegisteredUser, shown through inheritance. This means that they gain all functions from RegisteredUser, plus some additional content. Both RegisteredUser and the Contributor are linked to their profile classes, and they are AccountProfile and ContributorProfile. They store personalised information. These profiles contain basic details. When a user becomes a Contributor, they also gain the ContributorProfile, which expands on the basic profile by adding contributor-specific attributes, they are, numberOfRecipeUploaded and setRecipeRate. Then the DistinguishedMember class inherits from Contributor, taking additional content such as rateRecipe, longFormArticle, mentorshipTool, expertiseTag, and featuredSection. The advanced capabilities of this class are shown in the DistinguishedMemberProfile. This class of the class diagram provides publishing long-form articles, supports contributors, displays specialisation areas, and highlights recipes.

This class diagram contains Aggregation and Composition relationships. Those classes are 'IngredientList and Recipe' (Aggregation), 'AppSetting and User' (Composition), 'Comment and RegisteredUser' (Composition), and 'Recipe and Contributor' (Composition). The 'searchFilter' class provides filtering options, including ingredient filters, cuisine preferences, and time range. Also, this system supports content rating through the Rating class, meal planning through the 'MealPlan' class, and the organisation of recipes through the 'ChefTips' class, and this provides preparation guidance linked to specific recipe sections. These components of the class diagram work together to create a proper class diagram.

Additionally, the above class diagram integrates five main design pattern diagrams and they belong to Decorator, Singleton, and Strategy patterns. These patterns were used to enhance the flexibility and maintainability. The Strategy pattern which was done by Marisha is applied to

optimize grocery lists using various processing methods. The Decorator Patterns which are done by Senesh, Kavindi, and Viysh are applied three times while the Singleton pattern which was created by Mevindi manages global applications settings to make sure consistent configuration across the platform. Doe to word count limitations, not every class in the class diagram are mentioned in detail. This explanation focuses on the most significant classes.

2. Question On State Machine Diagram

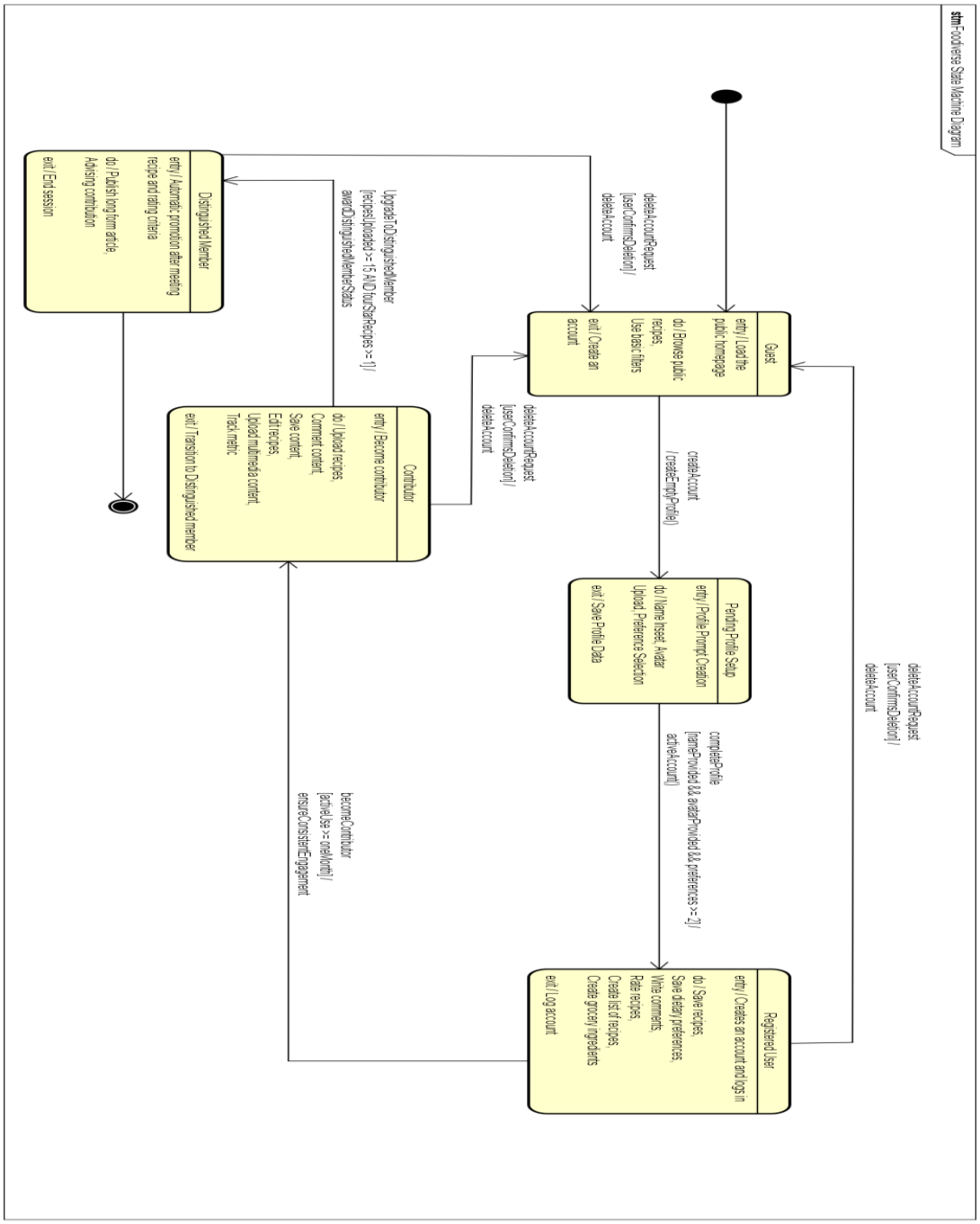

Figure 2 State Machine Diagram

2.1. State Machine Diagram Explanation (300 words)

A state machine diagram captures all the possible responses of a single object to all the use cases in which it is involved. The above-mentioned state machine diagram shows the user journey levels of the "Foodiverse" global recipe sharing platform. The above state machine diagram contains six states during the user journey lifecycle, and they are the Initial pseudo state, the Guest state, the Registered User State, the Contributor State, the Distinguished Member state, and the final state. Every state of this state machine diagram consists of Internal transitions, and they are Entry Activity, Do Activity, and Exit Activity. Plus, all transitions of this state machine diagram are labelled with 'Trigger, Guard, and Activity'.

1st Transition - This happens from the initial state to the Guest state. During this transition, the trigger, guard, and activity can't be found.

Guest State - In here, the initial state becomes the Guest state. The Guest state contains internal transitions in the Entry level: Load the public homepage, in Do level: Browse public recipes and use basic filters, in Exit level: Create an account.

2nd Transition and 2nd State - Guest transitions to the state of Pending Profile Setup. To make this transition, the Trigger: create account, the Guard: No Guard conditions, the Activity: create empty profile. Inside the Pending Profile Setup state, there are some internal transitions that happen. In Entry level: Profile Prompt Creation. In Do level: Name Insert, Avatar Upload, and Two preference selection. In Exit level: Save Profile Data.

3rd Transition and 3rd State – Pending Profile Setup transitions to the Registered User state. To make this transition, the Trigger: Complete Profile, The Guard: Name should be provided, Avatar should be provided, and the two preferences should be selected, The Activity: Active Account. Inside the Registered user state, there are some internal transitions that happen. In Entry level: Create and account and logs in, In Do level: Save recipes, Save dietary preferences, Write comments, Rate recipes, Create list of recipes, and Create grocery ingredients, In Exit level: Log account.

The Above-mentioned Guest state to the Pending Profile Setup state transition and the Pending Profile Setup state to the Registered User state transitions are user driven not automatically. If a user cannot function in the next state until some conditions are fulfilled, we use this transition. Because of this reason the state must be its own state, not an entry action.

4th Transition and 4th State - Registered user to the state of Contributor. To make this transition, the Trigger: becomes a contributor, the Guard: maintains consistent engagement with the platform for at least one month through active use, and the Activity: ensures consistent engagement for one month. Inside the Contributor state, there are three types of transitions that happen. In Entry level: Become a contributor, in Do level: upload recipes, comment content, save content, edit recipes, upload multimedia content, and track metrics, in Exit level: Transition to Distinguished member from Contributor state.

5th Transition and 5th State - This happens from the Contributor state to the Distinguished Member state. To make this transition, the Trigger: Upgrade to the distinguished member, the Guard: upload more than 15 recipes and have at least one recipe rated 4 stars or higher by other users, the Activity: transition to the Distinguished member status. Inside the Distinguished member state, internal transitions happen. In Entry level: Automatic promotion after meeting the recipe and rating criteria, in Do level: Publish long form article and advising contribution, in Exit level: End of the session. After this state, the lifecycle transitions to the Final state of the state machine diagram.

Other main transition - If the Registered user, Contributor, and the Distinguished member want to delete their account, there is a transition for that also. From that transition, the above-mentioned states transition from their states to the Guest state again. To make this transition, the Trigger: request to delete their account, the Guard: confirmation of deletion, and the Activity: deleting accounts. From this transition, the above-mentioned users can delete their accounts without meeting the final state of the lifecycle of this state machine diagram.

2.2. Appendix (Other Way of The State Machine Diagram)

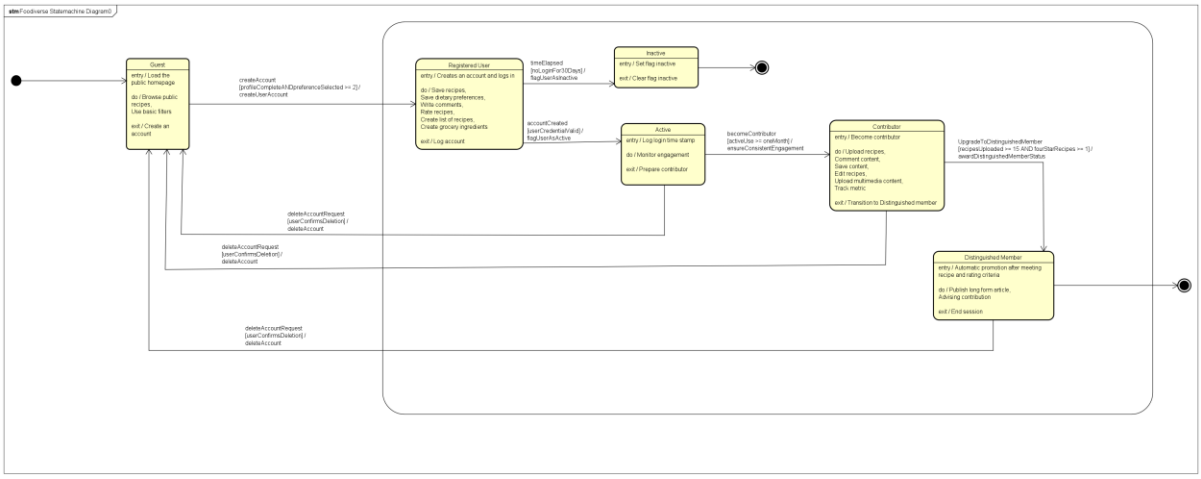

Figure 3 Appendix State Machine

- In this State Machine Diagram the 2nd Transitions and 2nd States - Registered User state to the Inactive and Active states. To make this transition from Registered user to Inactive state, the Trigger: time elapsed, the Guard: not login for 30 days, the Activity: Flag this user as Inactive. To make the transition of Registered User to Active state, the Trigger: account created, the Guard: Valid user credential, and the Activity: Flag this user as active. In here the state machine diagram becomes a Nested state machine. 3rd Transitions and 3rd states - Inactive User to final state, Inside the Inactive state, there are two internal transitions happen. In Entry level: Select flag inactive, in Exit level: Clear flag inactive. The transition happens from the Inactive state to final state. The above transitions the Inactive and Active states are concurrent states. They are concurrent sub activities. When the person who became the registered user face a transition to the Inactive state and from inactive state to the final state because the Registered user has not completed the 1 month consist engagement. In the scenario they didn't mentioned that the registered user who didn't complete his consist engagement becomes a Guest again. 4th Transition and 4th State - active state to the state of Contributor. To make this transition, the Trigger: becomes a contributor, the Guard: maintains consistent engagement with the platform for at least one month through active use, and the Activity: ensures consistent engagement for one month. Inside the Contributor state, there are three types of transitions that happen. In Entry level: Become a contributor, in Do level: upload recipes, comment content, save content, edit recipes, upload multimedia content, and track metrics, in Exit level: Transition to Distinguished member from Contributor state.

3. Question on Patterns - Identified Design Patterns and Responsible Group Members

I. Optimization Grocery Lists Using Different Methods (Strategy) – **Marisha**

II. Adding Optional Features to Registered Profile (Decorator) – **Kavindi**

III. Adding Optional Features to Distinguished Member Profiles (Decorator) – **Senesh**

IV. Managing Global Application Settings (Singleton) – **Mevindi**

V. Recipe Enhancement optional additions (Images, Nutritional Breakdown, Video Walkthrough, and ChefTips) to mandatory Recipe Component (Decorator) - **Vyishnavan**

4. Question On Graphical User Interface Design: Recipe Upload Page for Foodiverse

a. Content Presentation

The recipe upload page of the Registered user should contain compulsory and optional enhancements while maintaining clearness and ease of use. The screen should display a clear page title which is “Registered User” in the header of the user interface followed by structured sections with visible labels and input fields of the user interface. Mandatory fields and the Optional Enhancements are the main fields which should be included inside the interface.

Mandatory Section: This section of the interface must include ingredients list, preparation time, cooking time, level of difficulty, regional information according to the given scenario of Foodiverse while other features of the interface of the registered user and they are Profile personalization, saved recipes, dietary preferences, header of the interface which includes store name, registered user, contributor, and distinguished member page names, and the profile icon of the registered user and when the user clicks on this icon the written comments, rated recipes, favorite recipes, grocery ingredients, achievements, social links, meal plan, and personal blog will be displayed. Preparation and the cooking times can be added as hours and minutes. When the user clicks on the difficulty level, the drop down menu will show the Easy, Medium, and Hard status of the recipe while the Regional Information will be displayed when the user clicks on it. The “NEXT” button will at the bottom of the page will direct to the “OPTIONAL ENHANCEMENTS” side of the page.

Optional Enhancements: This comprises the Images of the recipe (Choose file section to select the desired image), Nutritional breakdown of the recipe, Video walkthrough of the recipe, and the “Chef Tips” section of the recipe. Plus, the “SAVE DRAFT” and the “UPLOAD RECIPE” action buttons provide control options. In the designed interface, it contains these things as Food Photos (with Choose file option), Video Guide URL (with Video URL paste option), Nutrition insights (with rich text editor), and Chef Tips (with rich text option).

b. Color Scheme Approach

Figure 4 Color Scheme

For this user interface design, out team was selected a warm, earth toned color palette using the Adobe Color tool software to make sure harmony and accessibility of the user interface. The Natural Base #F2DCC2, which is Warm Cream, was used as the background across all three screens, offering a soft alternative to stark white. This reduces eye strain and provides a welcoming canvas for the user. For the text and structure of the interface, our team applied Secondary #401B06, which is Dark Brown. This color ensures strong readability against the cream base while controlling warmth compared to pure black. The Primary #D95F18 code, which is Bold Orange, was kept for major impact elements such as the main title "FOODIVERSE" and critical buttons like "SAVE DRAFT" and "UPLOAD RECIPE", naturally guiding user attention through the interface. Accent colors add subtle context, which is Olive Green #B2BF4B borders input areas to call up freshness and health, while warm yellow #F2B84B highlights the "Chef Tips" icon. Together, these color palette choices support clarity, accessibility, and a friendly food-centric identity.

c. Design Standards and Conventions

(Visibility of the system status , User control and freedom, Consistency and Standards , Recognition Rather Than Recall, Aesthetic and Minimalist Design, Flexibility and Efficiency of Use)

The interface design of Foodiverse is strict with six proven usability heuristics that can be used to guarantee maximum efficiency. Visibility of the system status is a prioritized workflow with a series of steps, where the mandatory phase of the information is distinctly marked by the compulsory phase as the active phase, as the NEXT button signifies progress, with the active registered user tab displaying the current location on command. The user control and freedom are also preserved as there are the buttons SAVE DRAFT and UPLOAD RECIPE, and the user can save his/her work without forced completion.

To achieve Consistency and Standards, the system uses patterns of interactions with the familiar indicators (easy, medium, and hard) in the dropdowns of the “Difficulty Level” and the “Regional Information” to provide predictable behaviour. Recognition rather than recall is attained by clearly marking all the input fields such as, ingredients List and Video Guide URL so that a user dose not have to memorize what information to enter in. The Aesthetic and Minimalist Design makes the visual noise much less intense through the systematization of the intricate inputs, i.e. the exact spinners of hours and minutes, into clear open spaces. Lastly is Flexibility and efficiency of use through the use of profile icon menu, as an accelerator to enable user to access instantly the menu of “Written Comments”, “Rated Recipes”, “Favourite Recipes”, “Grocery Ingredients”, “Achievements”, “Social Links”, “Meal Plans”, and “Personal Blog”

d. Images of The GUI Design

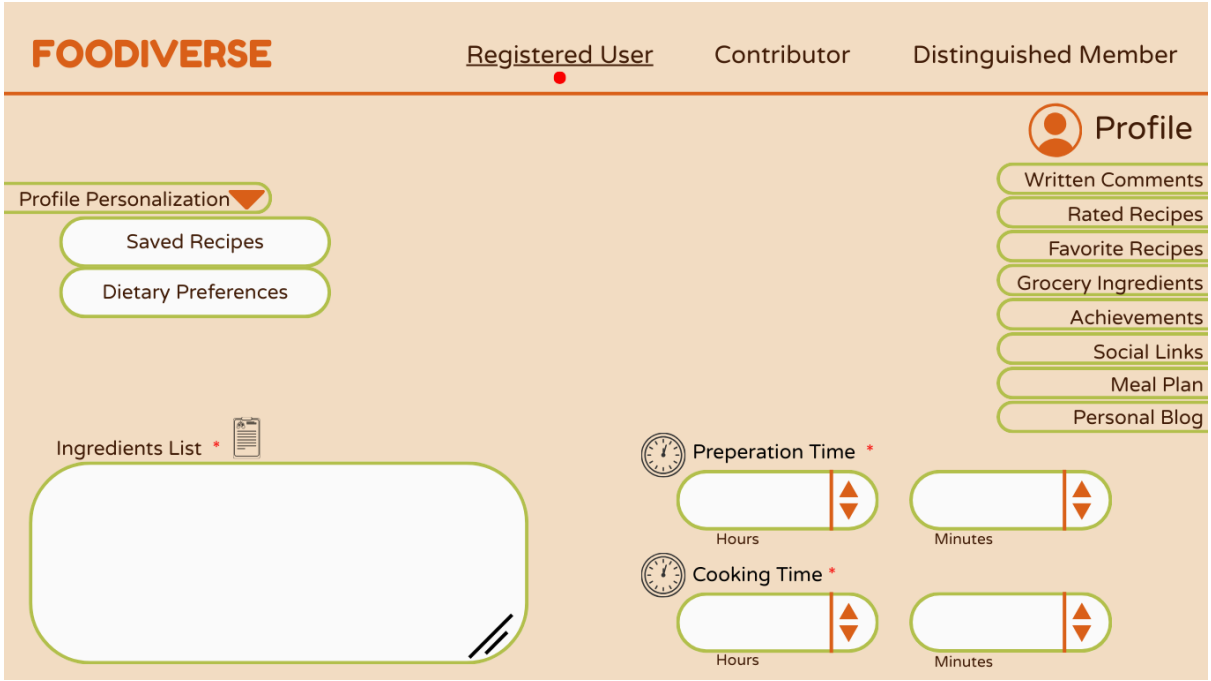

Figure 5 1st page of GUI

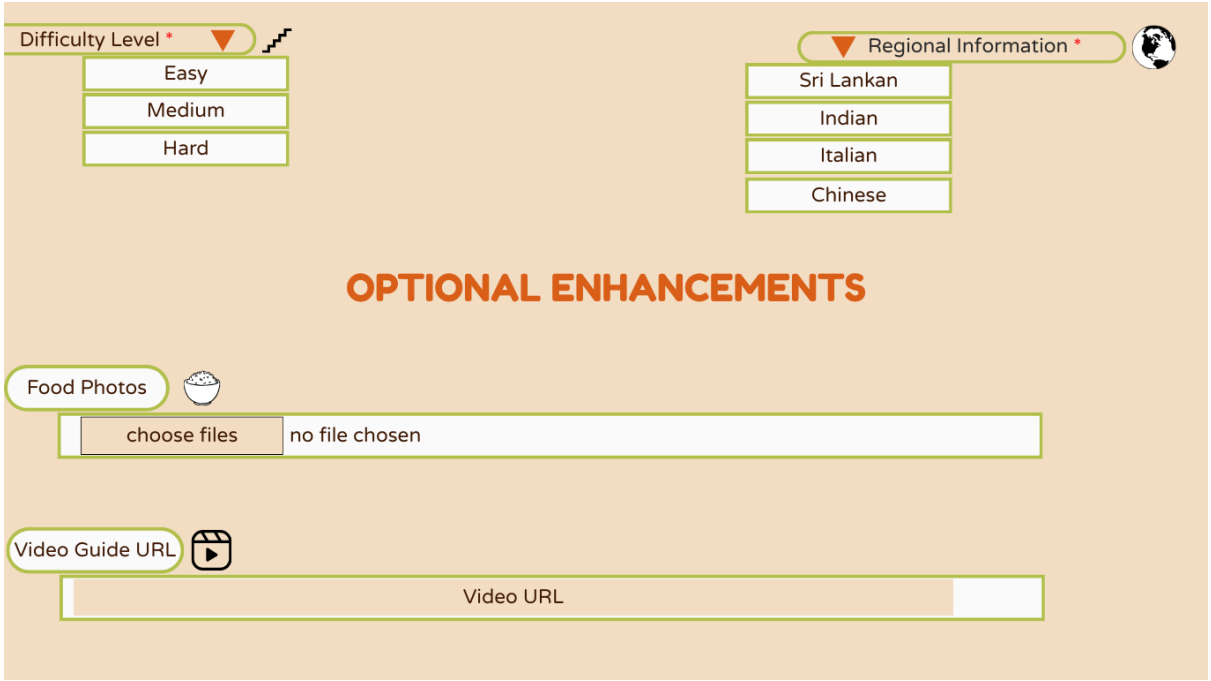

Figure 6 2nd Page of GUI

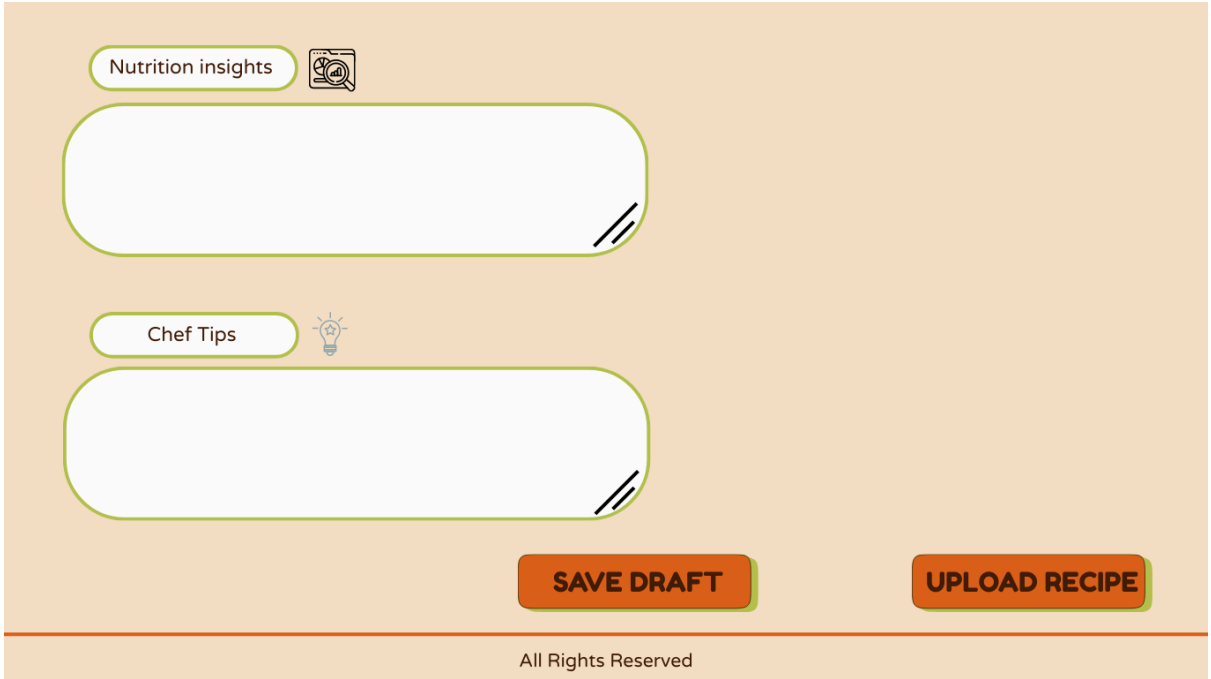

Figure 7 3rd page of GUI

e. References

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5. Record on Group Collaboration – 200 Words Paragraph

Meeting One – The initial class diagram was drafted from group coursework specifications and lecture material of level 4 and level 5. Each member of the group tasked with producing individual class diagrams for comparison and merging at the end of task one. The state machine diagram began individually, to be adjusted later. During this meeting the group completed the demo version of the class diagram, and the deliverables were distributed. This group meeting was the initial setting up of the coursework. Meeting Two – The members of the group reviewed the class diagram and discussed suitable design patterns and analyzed representation of user types. Patterns were selected under Decorator, Strategy, and Singleton design patterns as core patterns. Each member of the group tasked with drawing their own design patterns before the next meeting. Asked group members to prepare their individual 300 words explanations for each design pattern. The action items are Demo versions of pattern diagrams, feedback exchange, and lecturer input. This meeting was focused on Design Patterns. Meeting Three (Final Meeting) - This meeting was held at IIT Dialog Building Cafeteria on 27th of November 2025. The group reviewed the coursework specification report and group report for accuracy test. Finalized integration of main five design patterns into the class diagram. The GUI Design Ideas were sketched collaboratively; the "Canva" was chosen for editing. Action items were rewrite class diagram description, prepare explanations, proofread. Another group meeting was planned on 2nd of December to finalize the GUI design and proofread reports before submission. This meeting was the Final Integration and GUI. Apart from these main three group meetings, several group meetings were held by our group throughout the coursework period. Some of them were mentioned in the group meeting reports. All members actively participated and committed to complete this group assignment.

*** End of The Group Coursework Report ***